

ECMT676: Review Session

6.3 - 6.5 in Wooldridge (2010)

Department of Economics, Texas A&M University

- 1 Specification Tests
 - Testing for Endogeneity
 - Testing Overidentifying Restrictions
 - Testing Functional Form
- 2 (Correlated) Random Coefficient Models
- 3 Difference-in-Difference Estimation



Outline

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Testing for Endogeneity

- Durbin-Wu-Hausman
- H_0 : all explanatory variables are exogenous v.s. H_a : H_0 is false
- idea: the difference between the 2SLS and OLS estimators
-

$$(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})'[(\hat{X}'\hat{X})^{-1} - (X'X)^{-1}]^-(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})/\hat{\sigma}_{OLS}^2$$

where we must use a g-inverse, except in the very unusual case that all elements of X are allowed to be endogenous under H_a .



Endogeneity: Regression-based (CFA)



$$y_1 = \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + u_1$$

and

$$y_2 = \mathbf{z} \pi_1 + v_2$$

- H_o : $\text{Cov}(y_2, u_1) = 0$ which is equivalent to $\text{Cov}(v_2, u_1) = E(v_2 u_1) = 0$
- It is also equivalent to test H_o : $\rho_1 = 0$ of $u_1 = \rho_1 v_2 + e_1$
- We can estimate $\hat{\rho}_1$ from $y_1 = \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + \rho_1 \hat{v}_2 + \epsilon$



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Testing Overidentifying Restrictions

- When we have more instruments than we need to identify an equation, we can test whether the additional instruments are valid in the sense that they are uncorrelated with u



$$y_1 = \mathbf{z}_1 \delta_1 + \mathbf{y}_2 \alpha_1 + u_1$$

where $\mathbf{z}_1$ is $1 \times L_1$ and $\mathbf{y}_2$ is $1 \times G_1$.

- $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2)$ where $\mathbf{z}_2$ is $1 \times L_2$ and $L_2 > G_1$.



Testing OIR Cont'd

- $H_o: E(\mathbf{z}'u) = 0$
- Hausman Test: Based on a quadratic form in the coefficient differences
- Sargan Test:

$$NR^2 \sim \chi^2_{L_2 - G_1}$$

where R^2 is the R -squared from $\hat{u}_1$ on $\mathbf{z}$.

- Note that the test does not tell us which IVs fail the exogeneity requirement



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Testing Functional Form

- Sometimes we need a test for detecting neglected nonlinearities in models
- H_0 : $E(u|\mathbf{x}) = 0$, i.e. $E(y|\mathbf{x}) = \mathbf{x}\beta$
- LM test:

$$NR^2 \sim \chi_3^2$$

where R^2 is the R -squared from $\hat{u}_i$ on $\mathbf{x}_i$, $\hat{y}_i^2$, $\hat{y}_i^3$, and $\hat{y}_i^4$.

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Random Coefficient Models

- Recall that in 4.3.3 (page 73),

$$\begin{aligned}y &= \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \gamma_1 q + \gamma_2 q x_k + v \\ &= \beta_0 + \beta_1 x_1 + \cdots + (\beta_k + \gamma_2 q) x_k + \gamma_1 q + v\end{aligned}$$

where q is unobservable. The above equation is sometimes called a **random coefficient model** because one of the slope coefficients is "random" — that is, it depends on individual-specific unobserved heterogeneity.

RC cont'd

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- If q and $\mathbf{x}$ are correlated, we can consistently estimate the β_j by OLS if we have a suitable proxy variable for q . ▶ Proxy Variables

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Correlated Random Coefficient Model

- Things will become more difficult if one or more explanatory variables are endogenous and there are no proxy variables that break the correlation between the unobservable variables and the endogenous variables.



$$y_1 = \eta_1 + \mathbf{z}_1 \delta_1 + a_1 y_2 + u_1$$

where $\mathbf{z}_1$ is $1 \times L_1$, y_2 is the endogenous variables, and a_1 is the coefficient on y_2 , an unobserved random variable.

- $a_1 = \alpha_1 + v_1$ where $\alpha_1 = E(a_1)$

CRC cont'd



$$\begin{aligned}y_1 &= \eta_1 + \mathbf{z}_1 \delta_1 + (\alpha_1 + v_1)y_2 + u_1 \\&= \eta_1 + \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + (v_1 y_2 + u_1) \\&= \eta_1 + \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + e_1\end{aligned}$$

- The above equation shows explicitly a constant coefficient, α_1 , on y_2 but also an interaction between the unobserved heterogeneity.

- 2SLS? $\text{Cov}(\mathbf{z}, e_1) \stackrel{?}{=} 0$



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$$\text{Cov}(\mathbf{z}, e_1) = \text{Cov}(\mathbf{z}, v_1 y_2 + u_1) = \text{Cov}(\mathbf{z}, u_1) + \text{Cov}(\mathbf{z}, v_1 y_2) \neq 0$$

even if $E(u_1|\mathbf{z}) = E(v_1|\mathbf{z}) = 0$.

- Wooldridge(2003) shows that a sufficient condition for the IV estimator is $\text{Cov}(v_1, y_2|\mathbf{z}) = \text{Cov}(v_1, y_2)$

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Difference-in-Difference Estimation

- $y = \beta_0 + \beta_1 dB + \delta_0 d2 + \delta_1 dB \cdot d2 + u$ where y is the outcome of interest. The dummy variable dB captures possible difference between the treatment and control groups prior to the policy change. The time period dummy, $d2$, captures aggregate factors that would cause changes in y even in the absence of a policy change.
- δ_1 has been labeled the **difference-in-difference** estimator

- Let ΔY_i be the change in the value of Y_i for the i^{th} individual over the policy change. Assuming that the binary treatment variable $d B$ is randomly assigned, the OLS estimator $\hat{\beta}_1$ of $\Delta Y_i = \beta_0 + \beta_1 d B_i + v_i$ is the DID estimator [Introduction to Econometrics (Stock and Watson), Chapter 11].



Aside: Proxy Variable (see 4.3.2 on p.67)

Omitted variable bias can be eliminated, or at least mitigated, if a proxy variable is available for the unobserved variable q . Two formal requirements for a proxy variable z :

- $E(y|1, x_1, \dots, x_k, q, z) = E(y|1, x_1, \dots, x_k, q)$ (4.25): z is irrelevant in explaining y , in a conditional mean sense, once x and q have been controlled for.
- $L(q|1, x_1, \dots, x_k, z) = L(q|1, z)$ (4.26): The correlation between the omitted variable q and each x_i be zero once we partial out z .

Aside: Proxy Variable (see 4.3.2 on p.67)

- Why does the proxy variable do the job? Here, the model has cross product term of q we need a slightly stronger condition:

$$E(q|1, x_1, \dots, x_k, z) = E(q|1, z) = \theta_1 z$$



$$\begin{aligned} E(y|\mathbf{x}, z) &= \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \gamma_1 E(q|\mathbf{x}, z) + \gamma_2 E(q x_k|\mathbf{x}, z) \\ &= \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \gamma_1 \theta_1 z + \gamma_2 \theta_1 x_k z \end{aligned}$$

▶ Back to CRC